

MouseBot: Edge AI Evasion Robot

A vision-guided robot car that flees a pursuing person

Arduino UNO Q · ECE 180 · Final Project Proposal

Team #4

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What We Are Proposing

An autonomous robot car (ELEGOO chassis, re-controlled by the Arduino UNO Q) that uses its camera to detect a pursuing person and **drive away to avoid being caught** — while steering clear of walls and obstacles. All perception and decision-making run on-device, no cloud.

Where the AI/ML runs (on the AUQ):

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A TFLite person-detection CNN executes **on the UNO Q Ubuntu MPU**, producing the pursuer's *bearing* and *proximity* from each camera frame.

An on-device evasion policy converts detections (fused with an ultrasonic range sensor) into escape headings, sent to the STM32 for real-time motor control.

Must haves

- On-device person detection on the MPU
- Bearing + proximity estimation
- Reactive escape steering (flee the pursuer)
- Ultrasonic wall/obstacle avoidance on

Nice to haves

- Corner-aware evasion (avoid self-trapping)
- Learned evasion policy (RL in sim → transfer)
- Multi-pursuer / target re-identification
- Live FPV / detection overlay stream

Project Plan — Gantt Chart (Week 2 → Week 5)

